

"Falcon 2.0T: System voltage low from worn alternator under high electrical load"

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Area: bulletins

1 Condition

Certain Falcon 2.0T vehicles may set DTC P0562 (System Voltage Low) when operating with multiple electrical accessories engaged, such as heated seats, rear defrost, and headlamps at idle. The instrument cluster may display a battery warning lamp and the customer may report dimming headlights or a no-start condition after extended high-load operation. Confirm the concern by reviewing freeze-frame data for P0562 and verifying that system voltage drops below the regulated threshold under load. Refer to Falcon Charging Service Manual :: System Voltage Codes for the full criteria associated with this code.

2 Cause

The root cause is premature brush and slip-ring wear inside alternator assembly ALT-8810, which reduces field current and limits charging output once the alternator warms under sustained load. As output falls, battery BAT-1200 is forced to supply accessory demand and discharges below the regulated setpoint, triggering P0562. Excessive ripple from the worn alternator can also accelerate degradation of BAT-1200 over time. See Falcon Charging Service Manual :: Alternator and Falcon Charging Service Manual :: Battery for component background.

3 Repair Procedure

Begin by performing the charging system diagnosis in Falcon Charging Service Manual :: Charging System Diagnosis Procedure to confirm low output is attributable to the alternator rather than a wiring or control fault. Replace alternator ALT-8810 following procedure PROC-ALT, ensuring the drive belt and tensioner are inspected before final installation. After replacement, load-test battery BAT-1200 and replace it per Falcon Charging Service Manual :: Battery Replacement Procedure if it fails the test or shows evidence of deep discharge. Clear P0562, run the vehicle under simulated high electrical load, and verify regulated voltage is restored.

4 Parts Information

Order alternator assembly ALT-8810 for all repairs under this bulletin. If battery BAT-1200 fails load testing during the repair, order it as a supplemental part; do not replace BAT-1200 preemptively unless test results warrant it. Review the Falcon Charging Service Manual :: Charging System Overview for torque values and connector identification before completing the parts return.